Medical Microbiology

Third stage- College of Dentistry

Lab. 13

Enterobacteriaceae

Escherichia coli

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Learning Objectives:

After completing this lab, you should be able to:

1-Describe the microscopic morphology of members of the Enterobacteriaceae family.

2-Explain the laboratory techniques used for the detection and identification of *E. coli*.

3-Identify the cultural characteristics of *E. coli* and the appropriate media used for its isolation.

4-Discuss the pathogenic classification and major patho-types of *E. coli*.

Enterobacteriaceae:

The Enterobacteriaceae are a large, heterogeneous group of **Gram-negative rods.**

Morphology and General Characteristics

Shaped: rod-shaped bacteria.

Gram stain: Gram-negative.

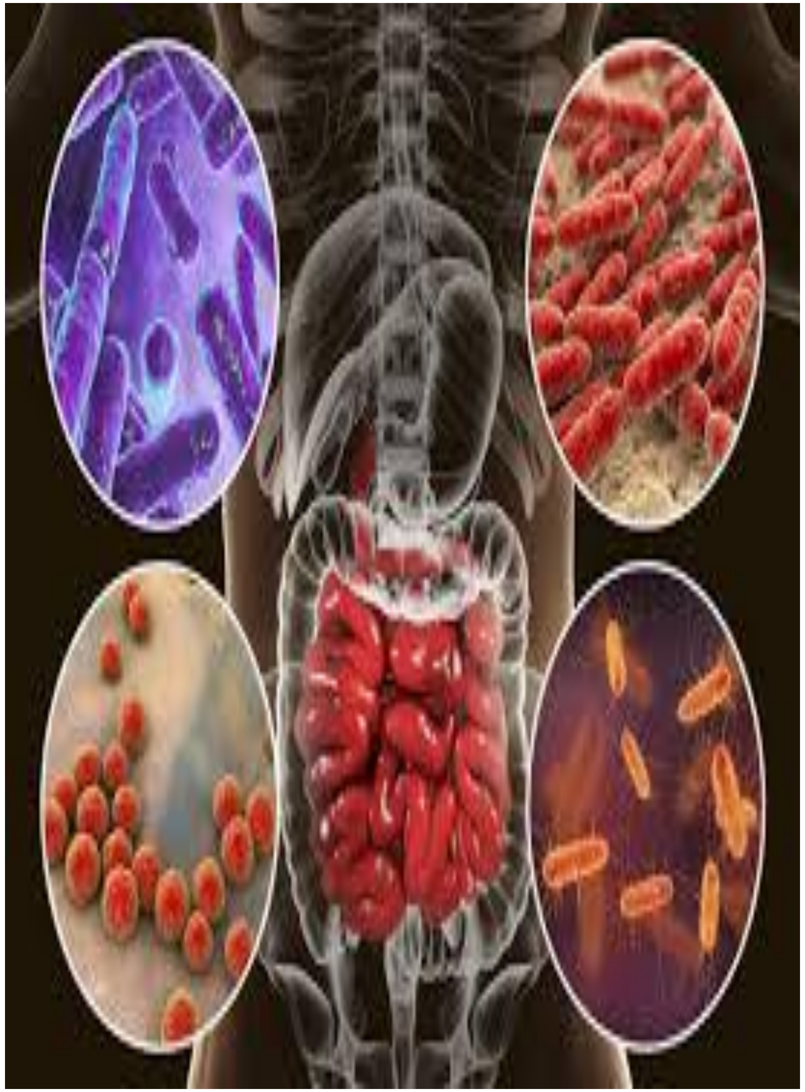
Respiration: Facultative anaerobe.

Spore formation: non-spore forming.

Motility: if motile -motility is by peritrichous flagella.

Capsule: Capsule may be found in some strains.

Habitat: natural habitat is the intestinal tract of humans and animals.



Enterobacteriaceae Classification

Enterobacteriaceae

Lactose fermenter

- *Echerichia coli*
- Klebsiella
- Enterobacter
- Serratia

Non Lactose fermenter

- Salmonella
- Shigella
- Proteus

Escherichia coli

General Characteristics

Shaped: rod-shaped,

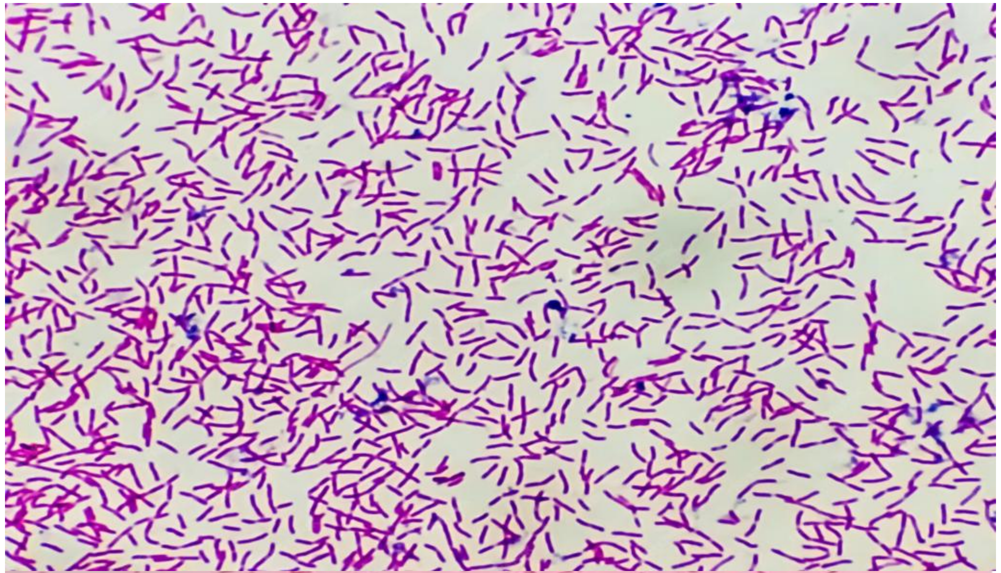
Gram-negative bacterium.

Natural Habitat: It lives naturally in the intestines of humans and warm-blooded animals as part of the normal intestinal flora.

Respirator: A facultative anaerobe

Biological Importance:

It plays a vital role in producing **Vitamin K** and preventing the colonization of pathogenic bacteria in the gut.



Pathogenicity and Virulence Factors

-Endotoxin

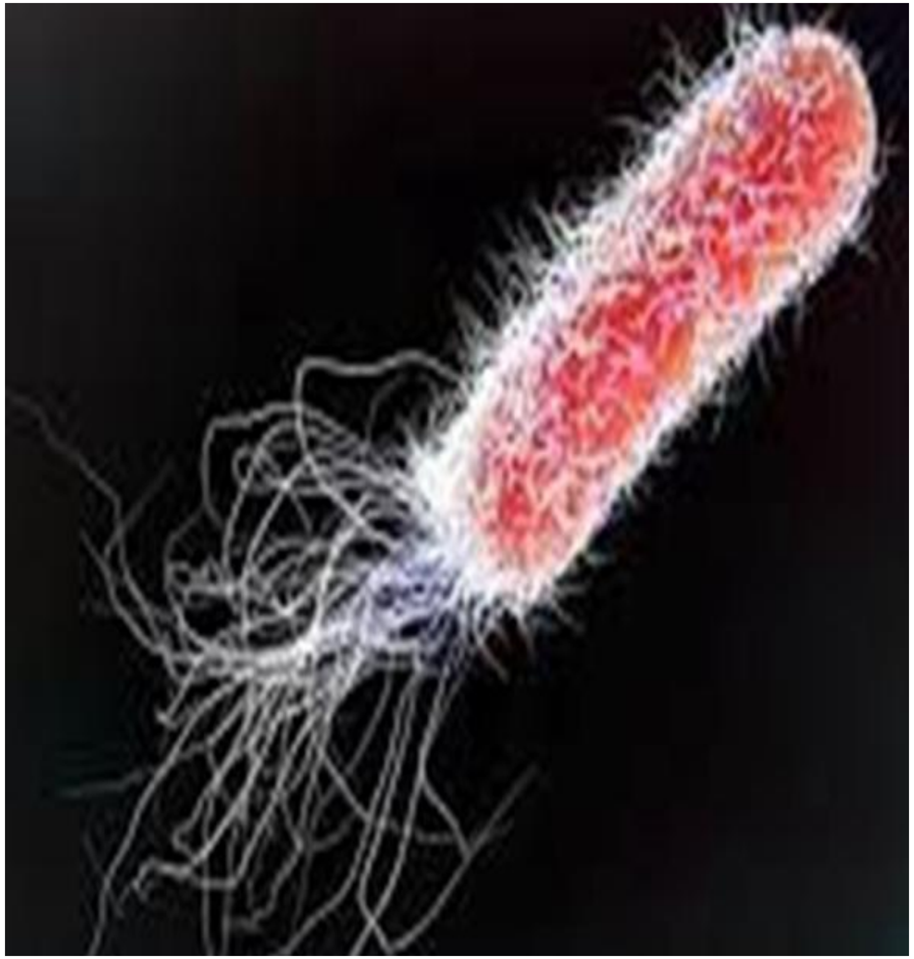
-Exotoxin-Enterotoxins, Cytotoxins.

-Hemolysins

-Capsule

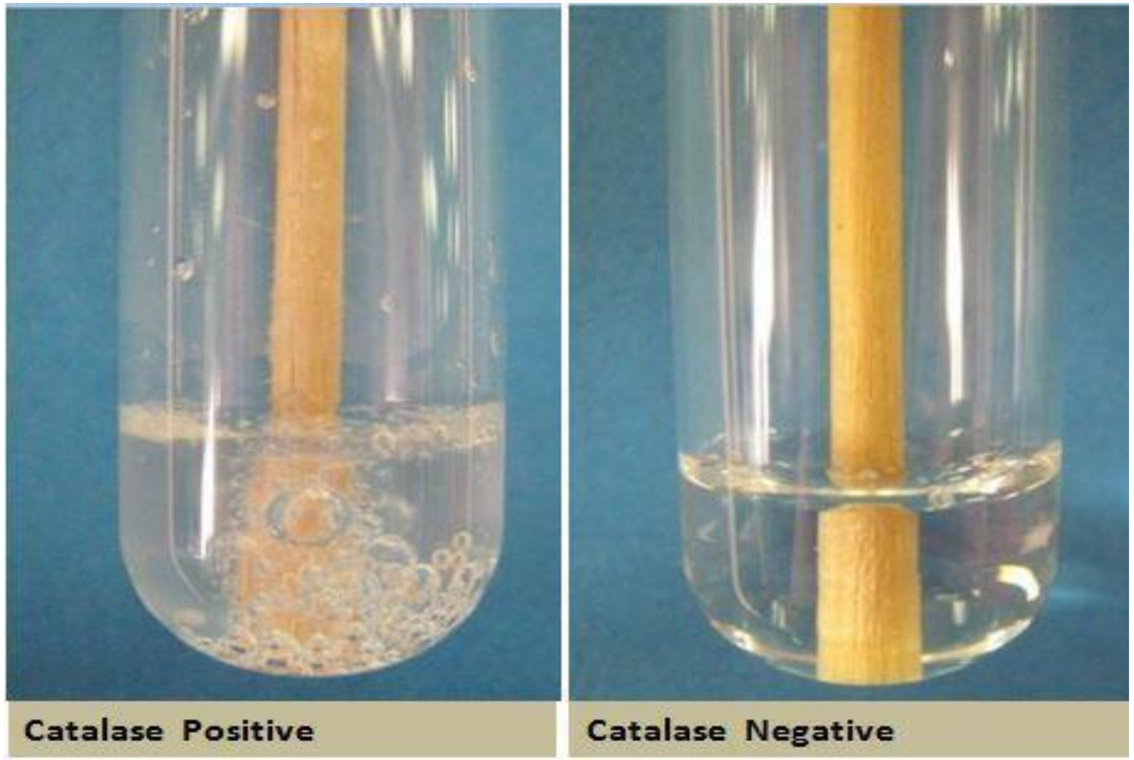
-Antigenic variation.

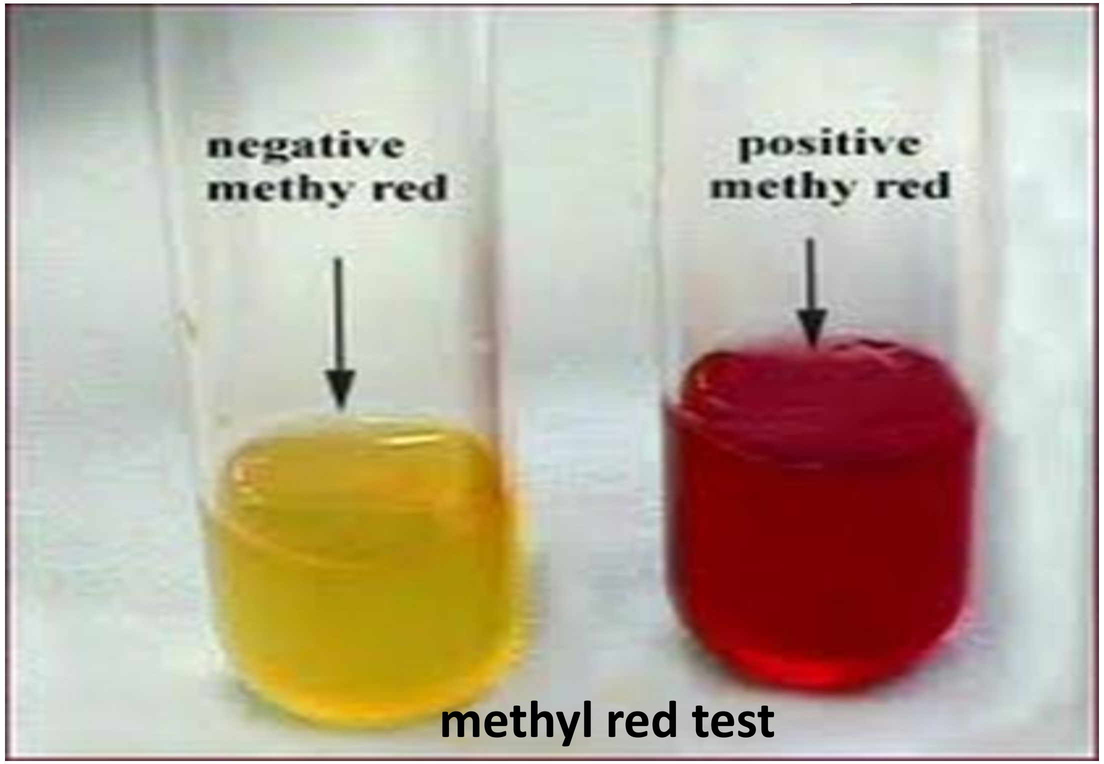
-Adhesins, fimbriae.



Culture media : Grows on MacConkey agar forms pink color colonies due to lactose fermentation.

Biochemical test	*E. coli*
Indole	Positive
Methyl red test	Positive
Urease	Negative
Catalase	Positive
Oxidase	Negative
Nitrate	Positive





Pathogenic Classification

While most strains are harmless, certain "pathogenic" strains cause serious diseases. They are classified based on their mechanism of infection:

1- Urinary tract infection (UTI):- The symptoms and signs include **urinary frequency**, **dysuria**, **hematuria**, and **pyuria**.

2. Sepsis:- when normal host defenses are inadequate, E. coli may reach the bloodstream and cause sepsis.

3. Meningitis:- E. coli and group B streptococci are the leading causes of meningitis in infant.

4. E. coli-associated diarrheal diseases:-

These E. coli are classified by the characteristics of their virulence properties.

1-Entero toxigenic E. coli (ETEC).

cause of “traveler's diarrhea”.

2-Entero pathogenic E. coli (EPEC).

Commonly associated with infant diarrhea.

3-Entero hemorrhagic E. coli (EHEC).

cause Hemolytic Uremic Syndrome leading to kidney failure.

4-Entero invasive E. coli (EIEC).

produce disease by invading intestinal mucosal epithelial cells.

5-Entero aggregative E. coli (EAEC).

causes acute and chronic diarrhea (>14 days in duration).

Any Question?

Thank you